

Assignment 2

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Q1:

.data

val1 DWORD 3

val2 DWORD 8

val3 DWORD 3

.code

main PROC

mov eax, val2

neg eax

mov ebx, val3

add eax, 7

sub eax, ebx

add eax, val1

INVOKE ExitProcess, 0

main ENDP

END main

Q2:

.data

array SBYTE 20, -5, 12

.code

main PROC

mov eax, 2

movzx ebx, array

sub eax, ebx

mov ecx, 7

movsx edx, array+1

sub ecx, edx

sub eax, ecx

movzx esi, array+2

add eax, esi

INVOKE ExitProcess, 0

main ENDP

END main

The screenshot shows a debugger window with the following components:

- Memory 1:** Address 0x00264004, displaying a memory dump of 20 bytes, all containing 0xFF.
- firstProgram.asm:** Assembly code listing:

```
5  
6 .data  
7 array SBYTE 20, -5, 12  
8  
9  
10 .code  
11 main PROC  
12 mov eax, 2  
13 movzx ebx, array  
14 sub eax, ebx  
15  
16 mov ecx, 7  
17 movsx edx, array+1  
18 sub ecx, edx  
19  
20 sub eax, ecx  
21  
22 movzx esi, array+2  
23 add eax, esi  
24  
25 INVOKE ExitProcess, 0
```
- Registers:**
 - EAX = FFFFFFFE
 - EBX = 00000014
 - ECX = 0000000C
 - EDX = FFFFFFFB
 - ESI = 0000000C
 - EDI = 00AF1005
 - EIP = 00AF1037
 - ESP = 006FFDAC
 - EBP = 006FFDB8
 - EFL = 00000286
- Status:** OV = 0 UP = 0, EI = 1 PL = 1, ZR = 0 AC = 0, PE = 1 CY = 0.
- Bottom Bar:** 150% zoom, No issues found, Call Stack, Breakpoints, Exception Settings, Command Window, Immediate Window, Output, Error List.